

# Quantitative Methods for Business Analysis

## Session 4: Conditional Risk, Expected Value, and Simulation



### Evaluating Uncertain Futures

Structure managerial decisions by updating conditional risks, weighting uncertain payoffs, and simulating full outcome ranges.

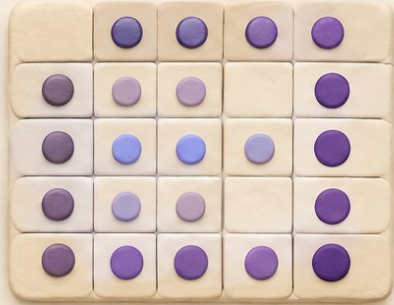
GC1015 Session 4 introduces foundational risk modeling frameworks, spanning probability updating to Monte Carlo simulation.

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# Navigating Managerial Uncertainty

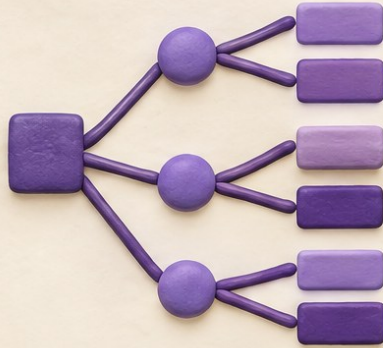
## From Diagnostic Signals to Full Distribution Modeling

### Probability Rules



Update operational risk estimates using contingency tables and diagnostic screening evidence.

### Decision Analysis



Map choices and uncertain demand states using payoff tables and backward induction.

### Monte Carlo Simulation



Overcome the Flaw of Averages by generating thousands of stochastic scenarios.

### Risk Workshop



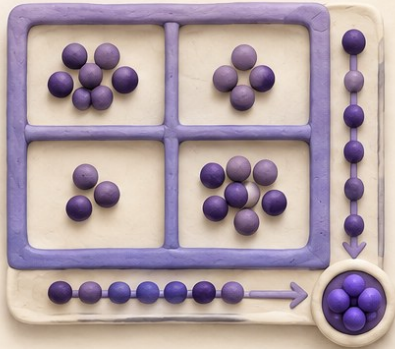
Evaluate capital trade-offs, conduct parameter sweeps, and audit dynamic formulas.

These four analytical methods build a rigorous bridge from single screening alerts to complete investment risk profiles.

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# Probability Rules and Conditional Risk

Interpreting Signals and Updating Baseline Beliefs



## Contingency Tables

Cross-tabulate categorical records. Margin sums give unconditional probabilities, while inner cells capture joint occurrences.



## The Double-Counting Fallacy

When events co-occur, simple addition overstates exposure. Always subtract the intersecting overlap.



## Bayesian Updating

Conditioning restricts the sample space. Diagnostic signals revise prior probabilities into actionable posterior risks.



## The Base Rate Effect

When base rates are low, false alarms can outnumber true detections. Do not confuse signal accuracy with risk.



**Rigorous probability screening provides the reliable likelihood foundations required for payoff tables and decision trees.**



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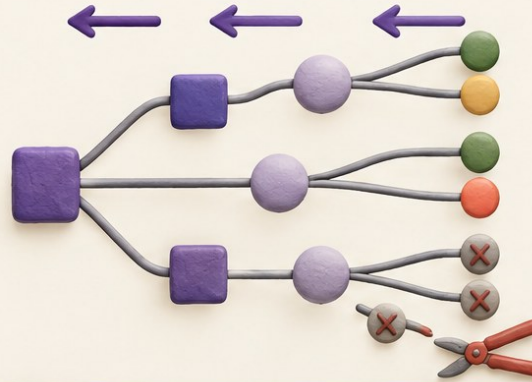
# Expected Monetary Value and Decision Analysis

## Structuring Sequential Choices Under Uncertainty



### Payoff Matrices

Map controllable choices against uncontrollable demand states. Expected monetary value weights each outcome by state probability.



### Rollback Induction

Solve decision trees backward from right to left. Average chance nodes and prune inferior alternatives at decision gates.



### Risk Attitudes

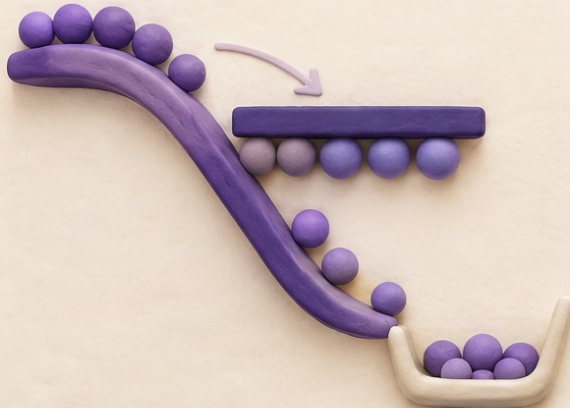
Expected value measures long-run averages. When downside losses threaten solvency, risk-averse managers require loss guardrails.

While decision trees structure discrete choices, continuous operational uncertainties require stochastic simulation.

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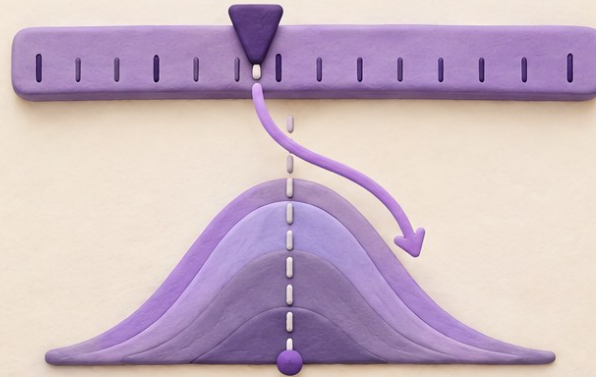
# Monte Carlo Simulation and the Flaw of Averages

## Modeling Non-Linear Payoffs Across Stochastic Distributions



### The Flaw of Averages

Plugging average inputs into non-linear models distorts expected profit whenever capacity caps upside and write-downs absorb downside.



### Generating Variates

Uniform random draws map through inverse transform functions to convert abstract probabilities into realistic operational inputs.



### Full Outcome Spread

Data tables recalculate dynamic models thousands of times to reveal Value at Risk and shortfall probability.

Simulation models provide the structural distributions necessary for stress-testing complex capital investment decisions.

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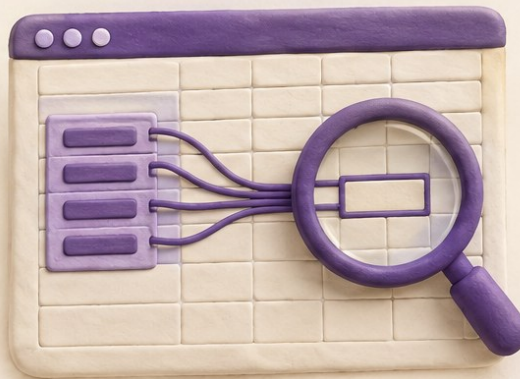
# Applied Risk Analysis and Simulation Workshop

## Auditing Capital Models and Communicating Downside Exposure



### Capital Trade-Offs

Compare high-capacity expansion against flexible modular options. Modest expected returns often shield firms from severe capital loss.



### Spreadsheet Auditing

Verify dynamic recalculation, ensure simulation tables reference blank workspace cells, and trace cell precedent formulas.



### Model Boundaries

More trials improve calculation precision, but cannot fix misspecified distributions or unmodeled multi-year market momentum.

**Rigorous formula auditing and transparent disclosure ensure quantitative risk distributions defensibly guide executive governance.**

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